INTRODUCTION TO MACHINE LEARNING

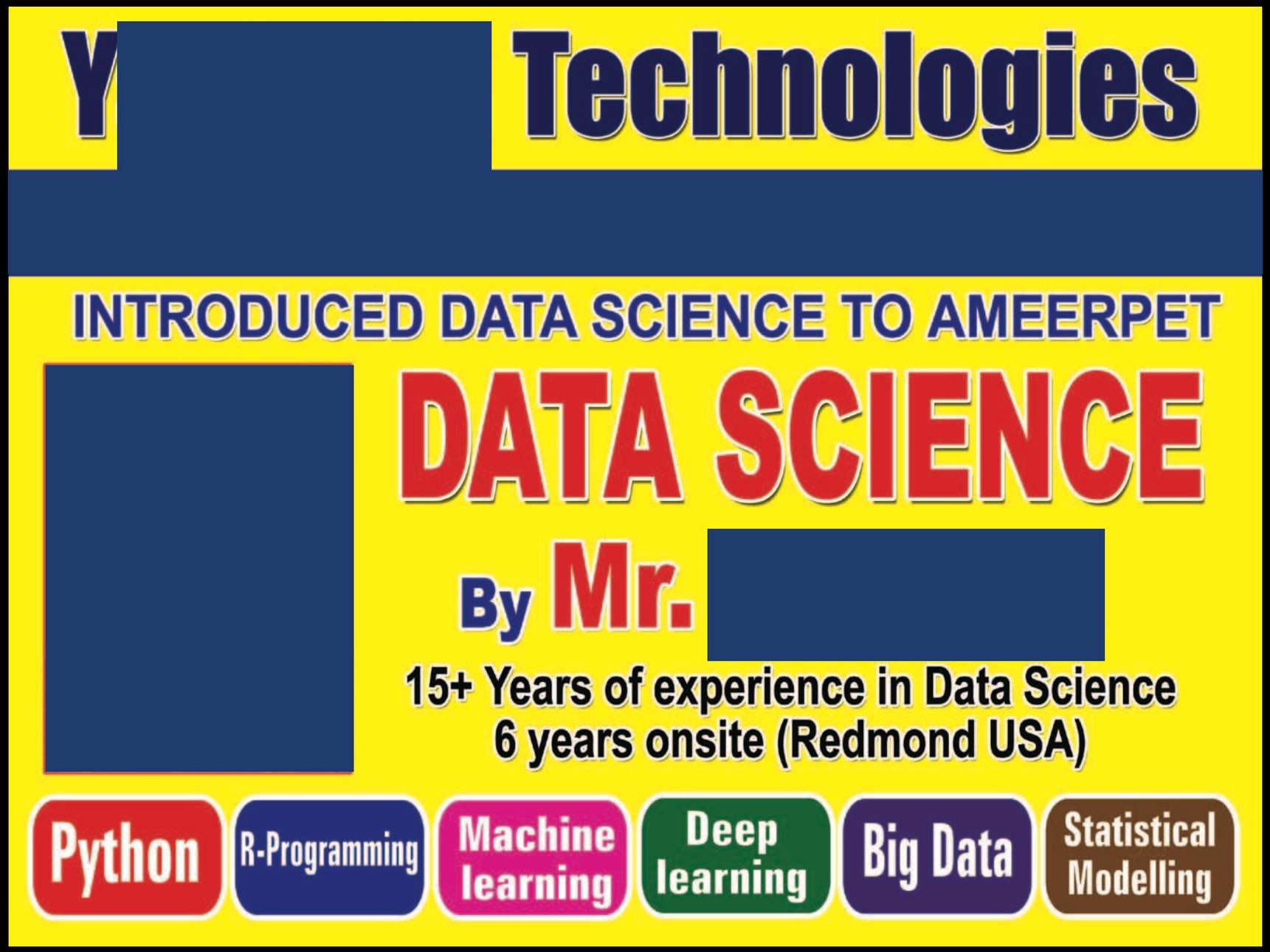
SUMOHANA S. CHANNAPPAYYA
IIT HYDERABAD



OUTLINE

- Motivation
- Machine Learning
- Supervised Learning
- Artificial Neural Networks
- Deep Neural Networks
- Current Trends

MOTIVATION


Technologies
INTRODUCED DATA SCIENCE TO AMEERPET
DATA SCIENCE
By Mr.
15+ Years of experience in Data Science
6 years onsite (Redmond USA)
Python
R-Programming
Machine learning
Deep learning
Big Data
Statistical Modelling

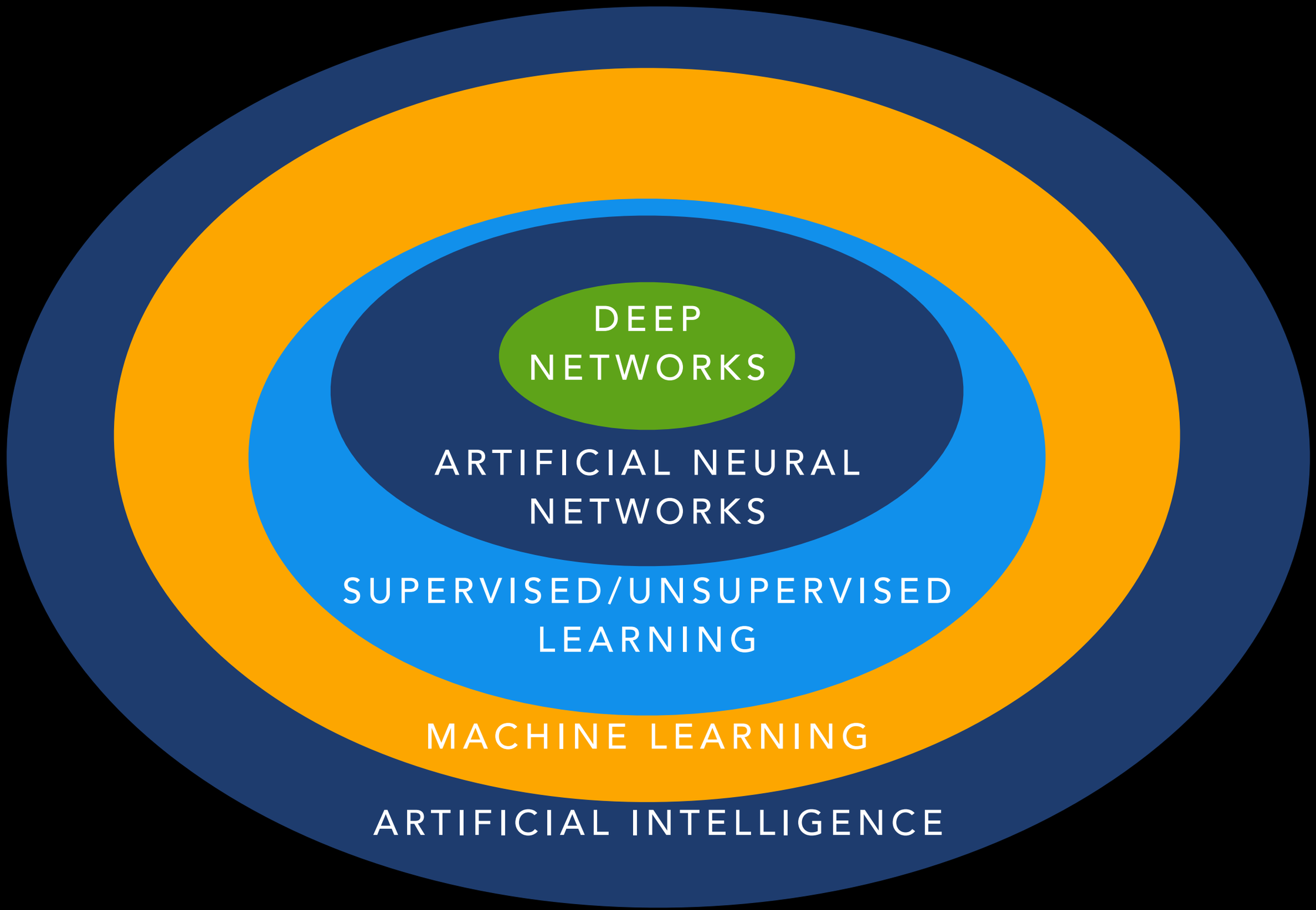
MOTIVATION


PG GENTS
EMINENT IT INFO
Weekday / Weekend Batch
DATA SCIENCE
R, STASTICS, PYTHON, MACHINE LEARNING
DEEP LEARNING, NLP, AI
Python
R Language

PERCEPTION OF DEEP LEARNING

ARTIFICIAL INTELLIGENCE = DEEP LEARNING!

DEEP LEARNING IN REALITY



BASIC MATH REVIEW

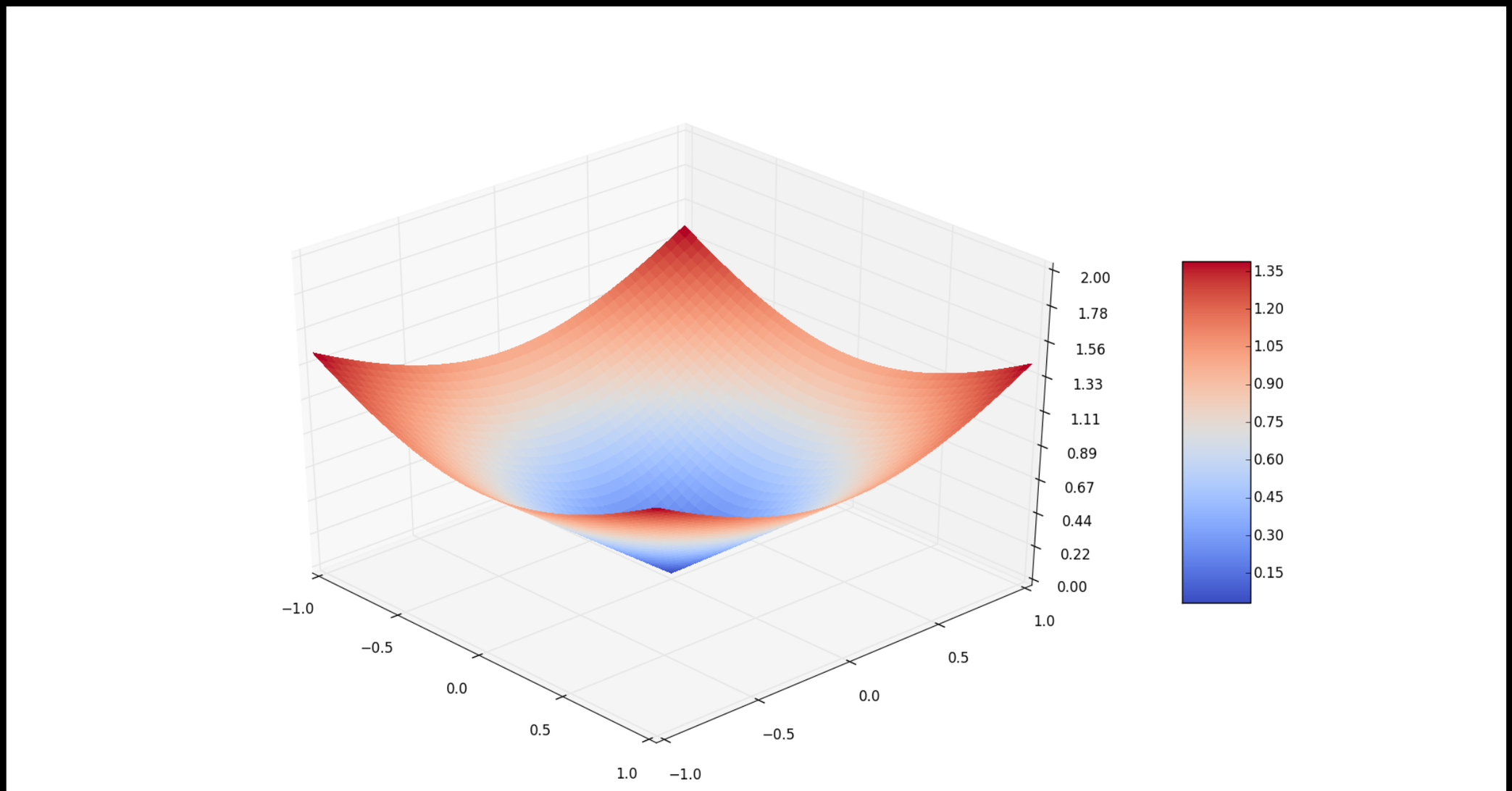
- A **vector** is denoted by: $\mathbf{x} = [x_1, x_2, \ldots, x_n]^T$
- Equation of a **straight line**: $\hat{y} = mx + c$
- Example of a **non-linear function**: $\sigma(x) = \dfrac{1}{1 + e^{-x}}$
- A **parameterised function** is denoted by: $\hat{y} = f(x; \theta)$

BASIC MATH REVIEW

- The **derivative/slope** of a function is zero at its maximum/minimum (if it exists): $\frac{dR(\theta)}{d\theta} = 0$
- The **gradient** of a multivariable function is like the **slope** of a single variable function

BASIC MATH REVIEW

- **Descending against the gradient takes us to the local-minimum**



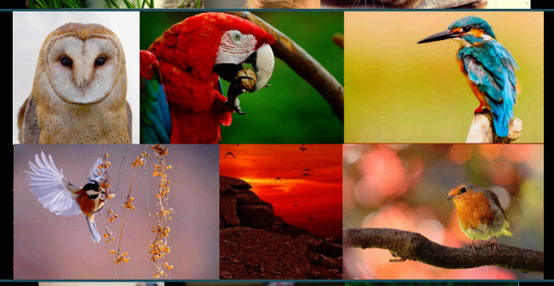
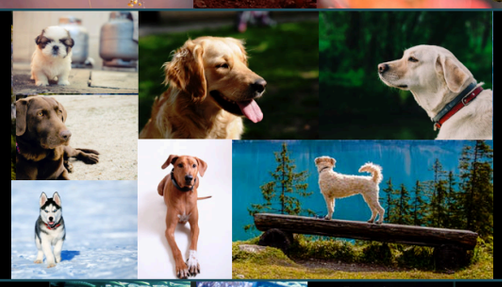
WHAT IS MACHINE LEARNING?

- **Learning from data:** a method of teaching machines to improve predictions from data
- A method for computers to **recognise patterns**
- "A computer program is said to learn from **experience E** with respect to some **class of tasks T** and **performance measure P** if its performance at tasks in **T**, as measured by **P**, improves with experience **E**."
  - *T. M. Mitchell, CMU*

MACHINE LEARNING FLAVOURS

- **Supervised Learning**
- Unsupervised Learning
- Reinforcement Learning

SUPERVISED LEARNING

Input (**x**)	Label (y)
	Cat
	Bird
	Dog
	Fish
	?

SUPERVISED LEARNING

- Given a set of **input points** and corresponding **ground truth labels:** $\{(\mathbf{x_1}, y_1), (\mathbf{x_2}, y_2), \ldots, (\mathbf{x_n}, y_n)\}$
- Discrete labels: **Classification**
- Continuous labels: **Regression**
- Learn a **parameterised functional relationship** to **estimate** label from input: $\hat{y} = f(\mathbf{x}; \theta)$

HOW DO WE LEARN PARAMETERS?

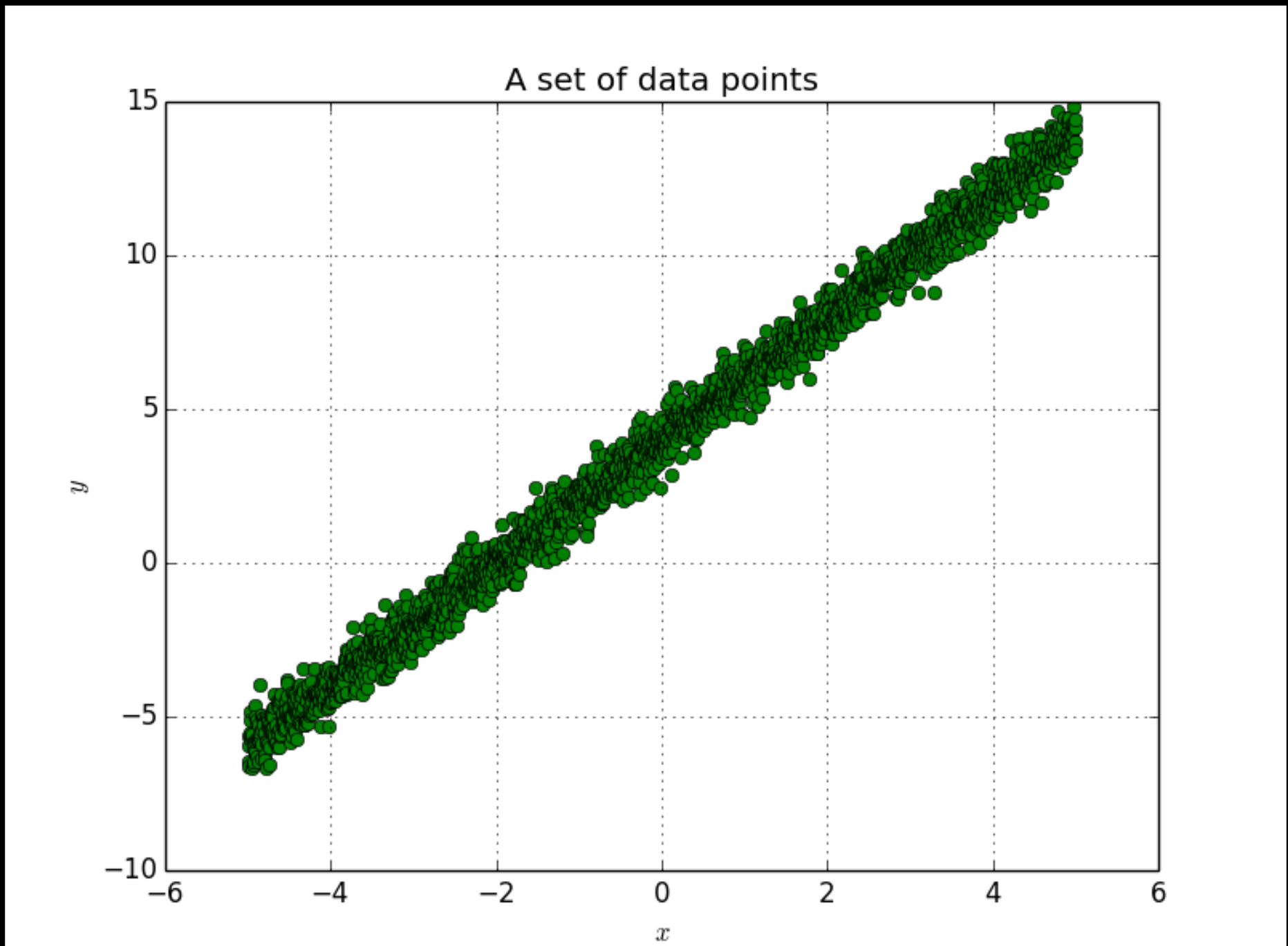
- How **good** is our **estimate** compared to **ground truth**? Quantify error: $d(y, \hat{y}) = (y - \hat{y})^2$

- Find **error** between **estimate** and **ground truth** over **all training data points**: $R(\theta) = \sum_{i=1}^{n} d(y_i, f(x_i; \theta))$

- Find **parameters** such that **error is optimised:** $\frac{\partial R(\theta)}{\partial \theta}$

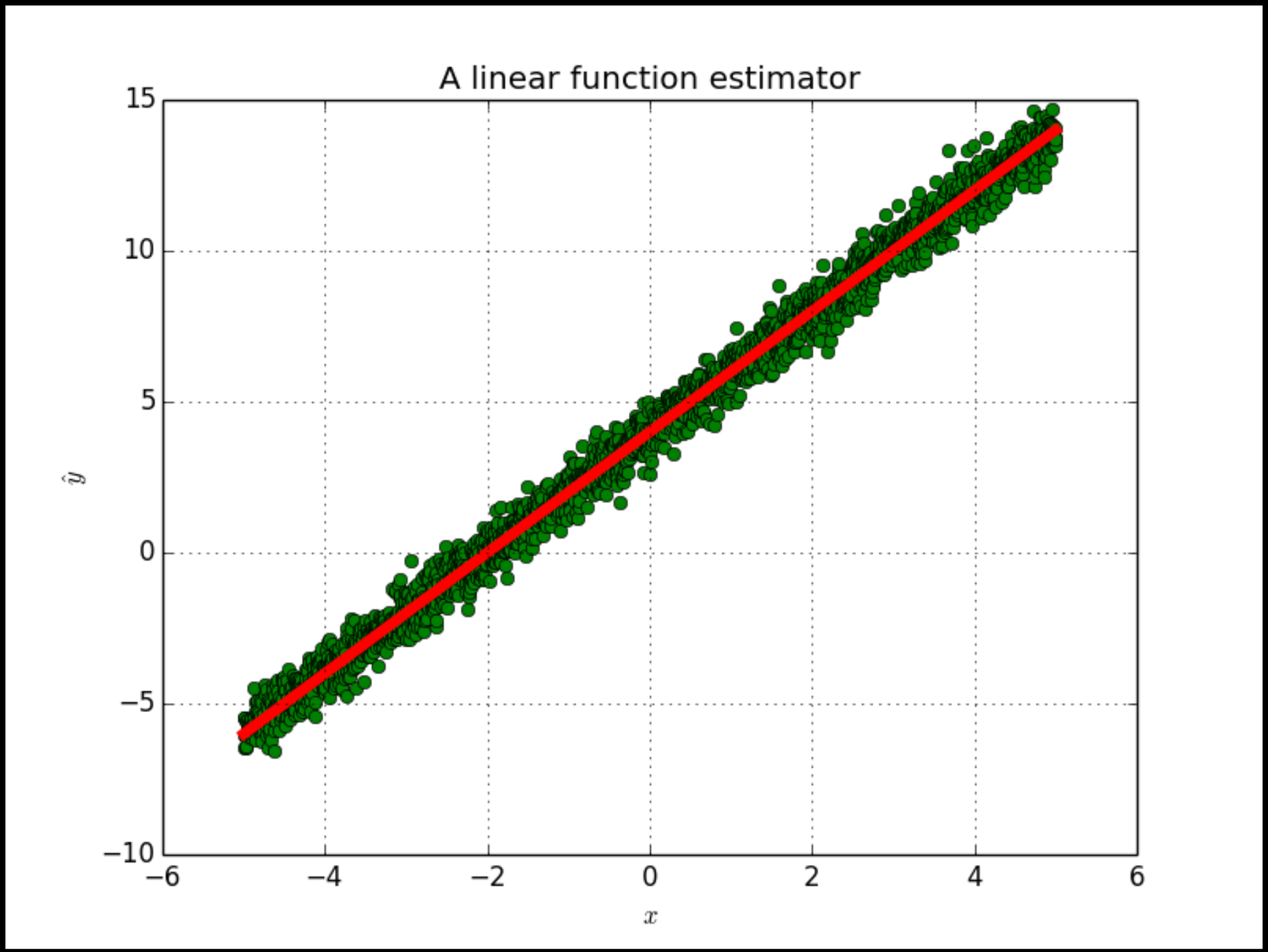
SUPERVISED LEARNING EXAMPLES

- Example 1:



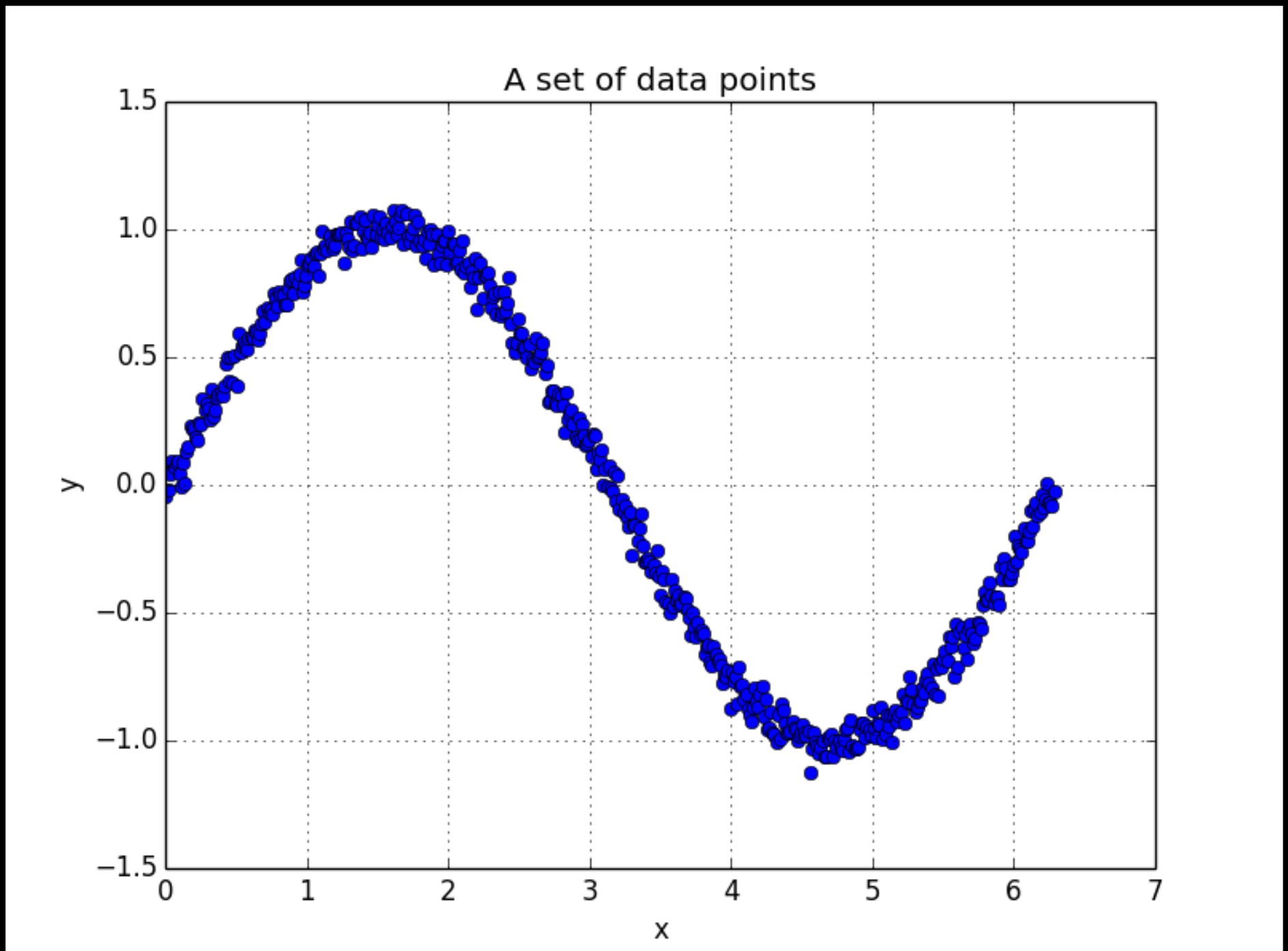
SUPERVISED LEARNING EXAMPLES

- Linear: $\hat{y} = mx + c; \theta = \{m, c\}$



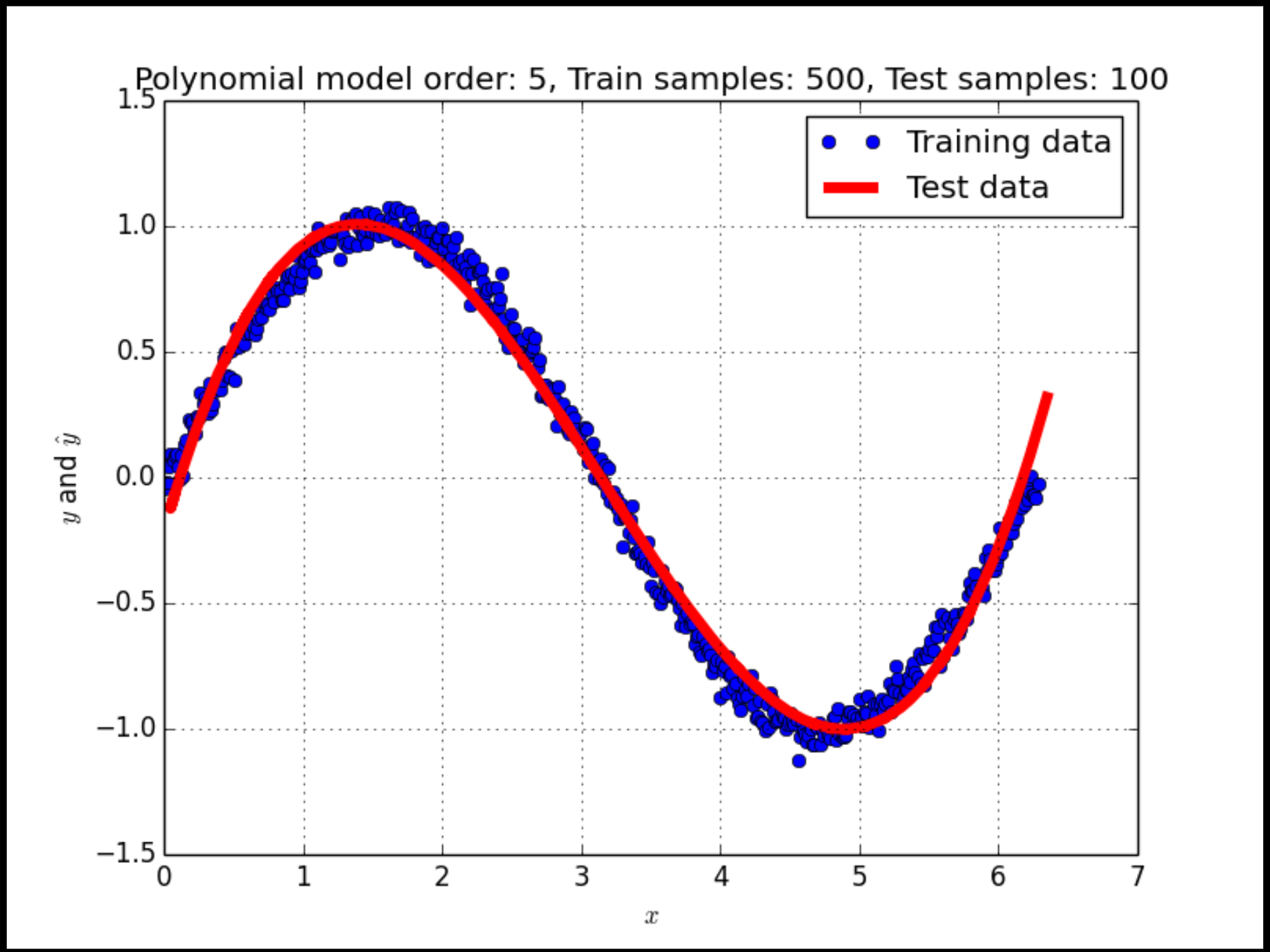
SUPERVISED LEARNING EXAMPLES

- Example 2:



SUPERVISED LEARNING EXAMPLES

- Linear: $\hat{y} = w_0 + w_1 x + w_2 x^2 \ldots + w_n x^n; \theta = \{w_0, w_1, \ldots, w_n\}$



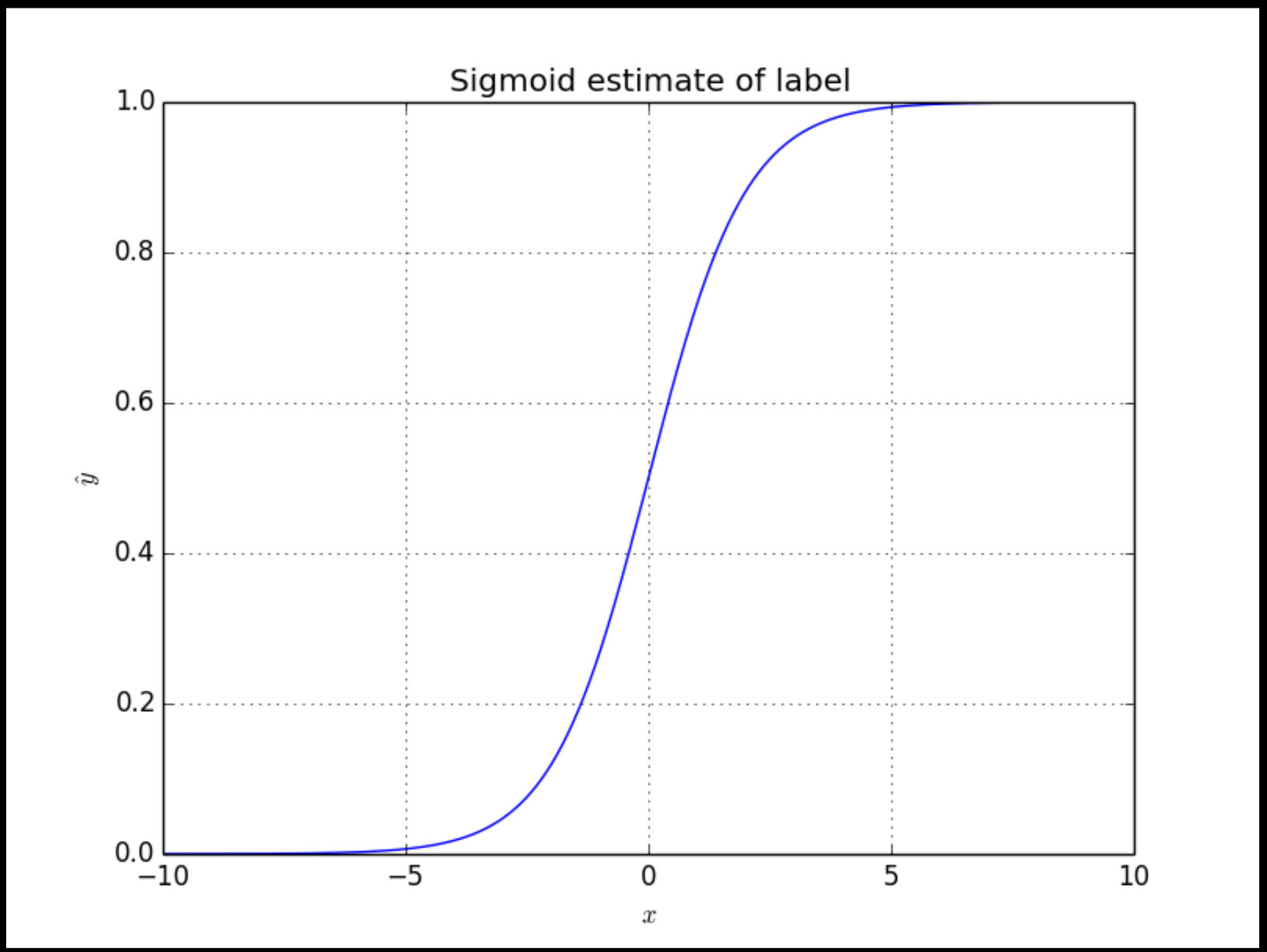
SUPERVISED LEARNING EXAMPLES

- Example 3:

Input (**x**)		Label (y)
0	0	0
0	1	1
1	0	1
1	1	0

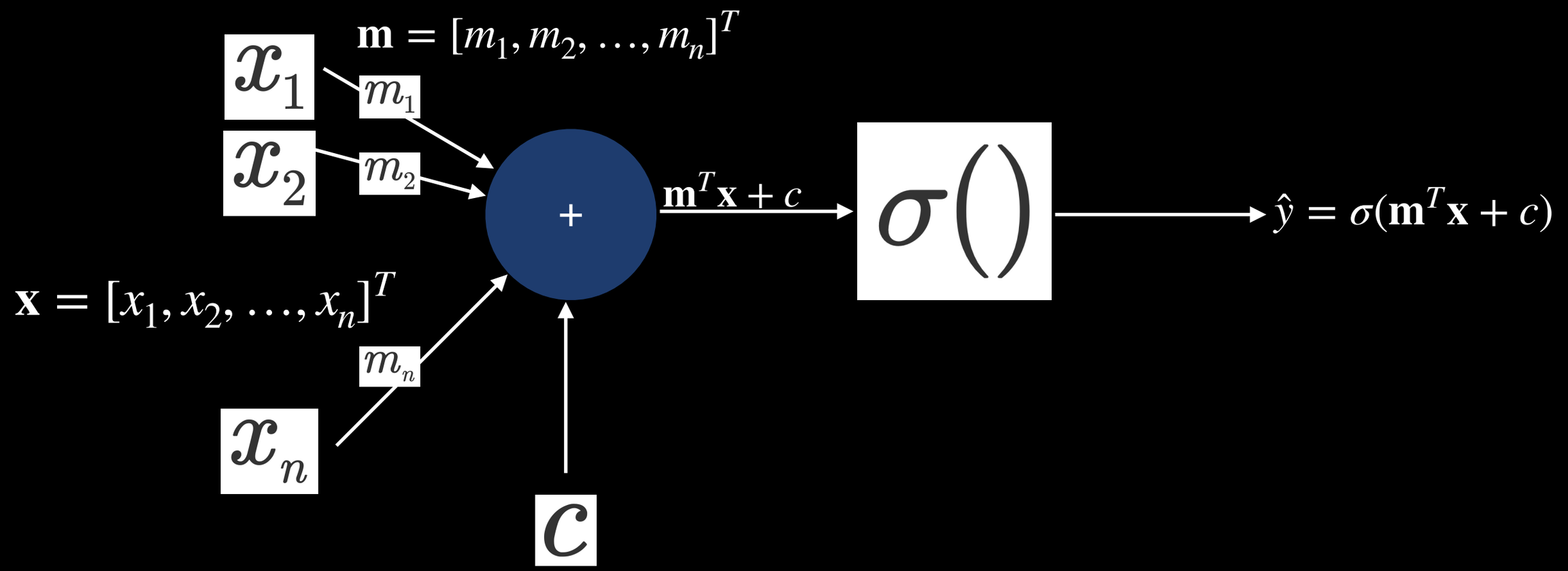
SUPERVISED LEARNING EXAMPLES

- Non-linear: $\hat{y} = \sigma(mx + c); \theta = \{m, c\}; \sigma(x) = \frac{1}{1 + e^{-x}}$



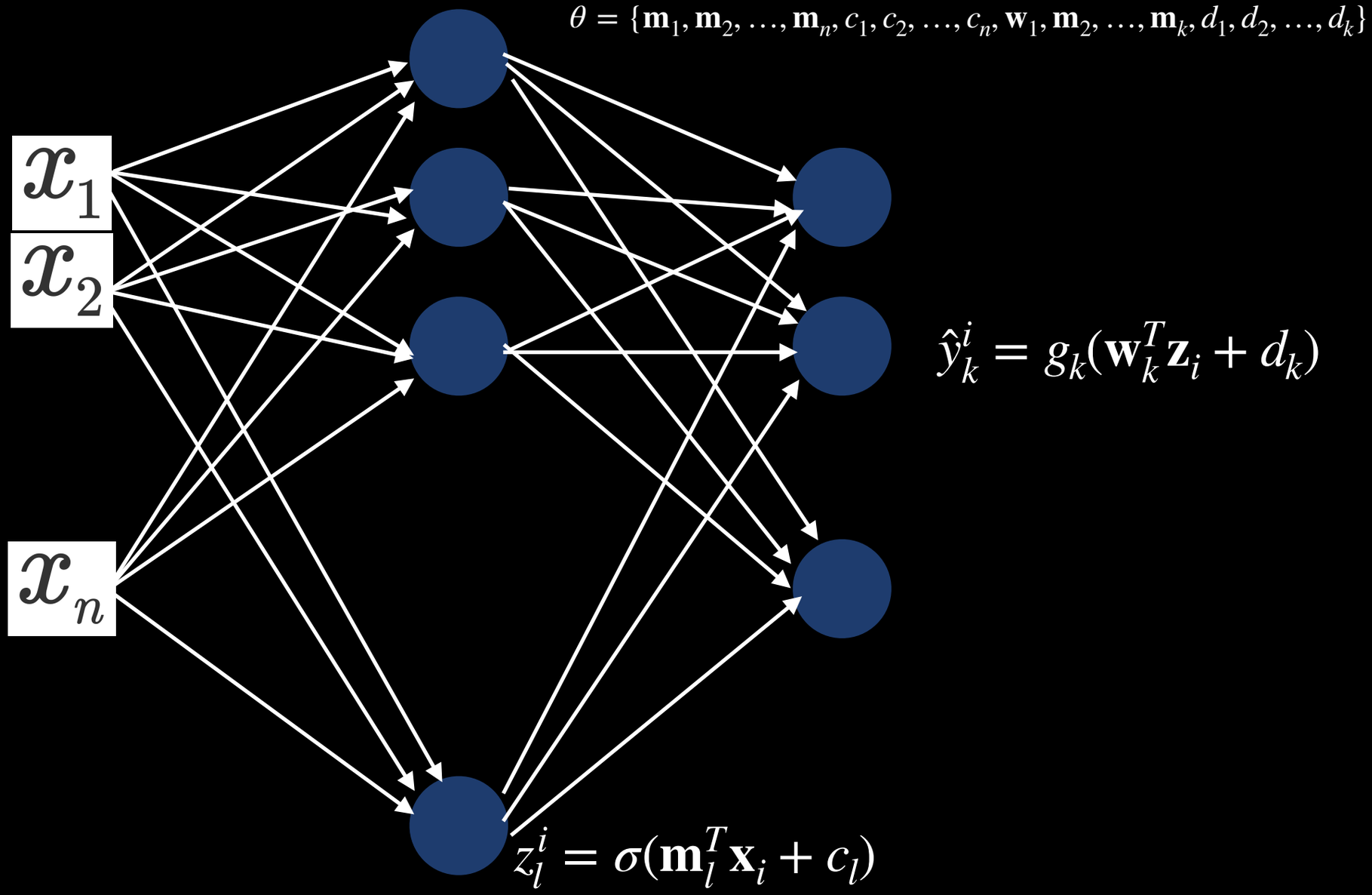
SUPERVISED LEARNING EXAMPLES

- Artificial Neuron: $\hat{y} = \sigma(\mathbf{m}^T\mathbf{x} + c); \theta = \{\mathbf{m}, c\}$



SUPERVISED LEARNING EXAMPLES

- Artificial Neural Network (ANN): $\hat{y}_k^i = g_k(\mathbf{w}_k^T \mathbf{z}_i + d_k) \quad z_l^i = \sigma(\mathbf{m}_l^T \mathbf{x}_i + c_l)$

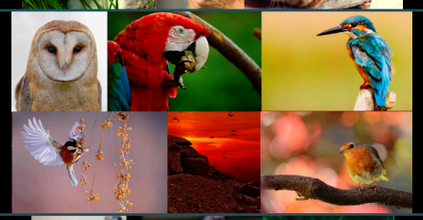
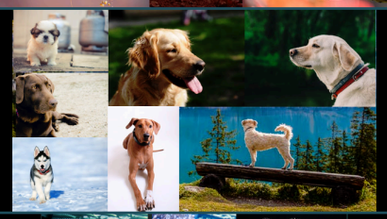


SUPERVISED LEARNING EXAMPLES

- ANN:
  - 2 input nodes
  - 2 hidden nodes
  - 1 output node
- Performance: very high accuracy of prediction!

SUPERVISED LEARNING EXAMPLES

- Example 4:

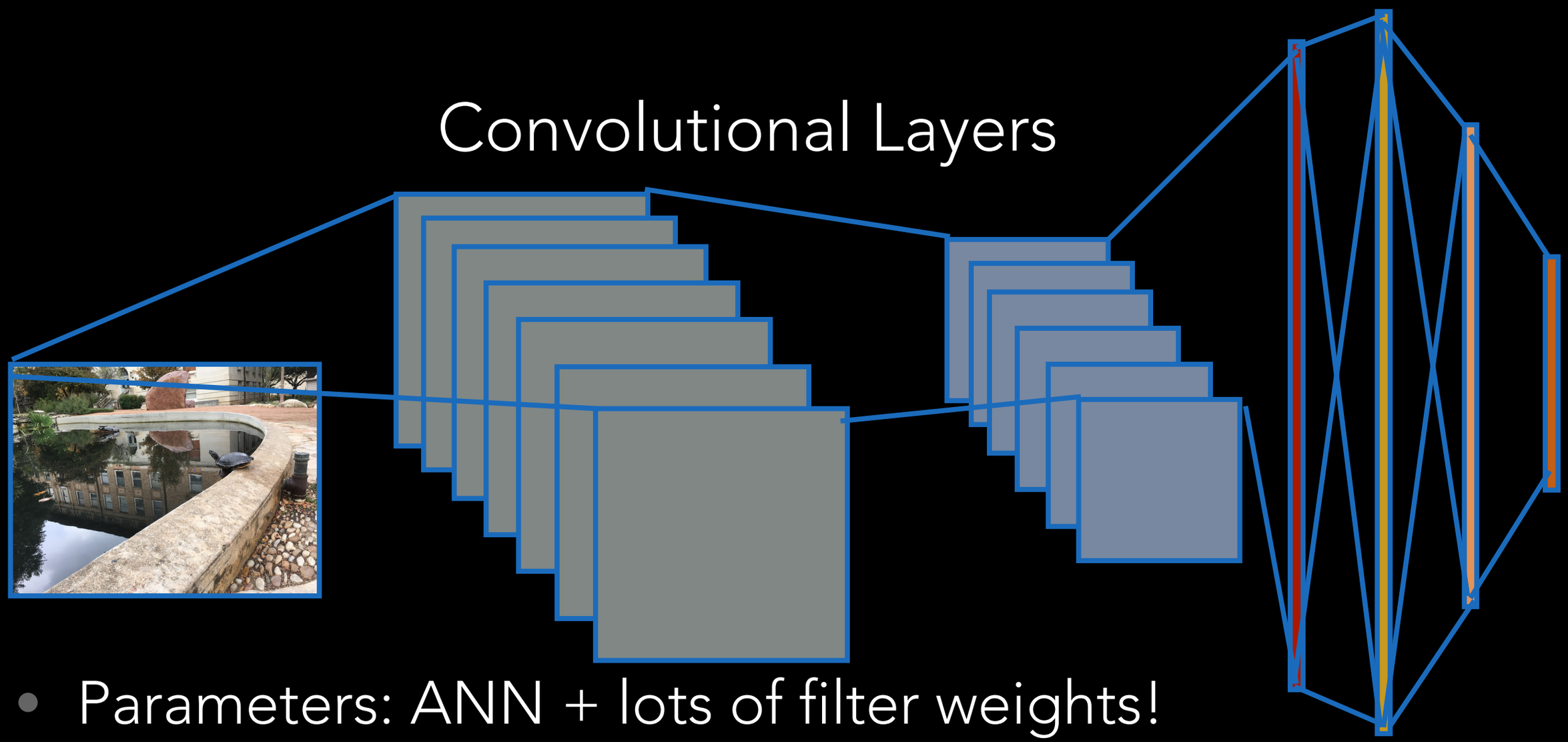
Input (**x**)	Label (y)
	Cat
	Bird
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	?

SUPERVISED LEARNING EXAMPLES

- Questions:
  - Can we **directly** relate 2D inputs to labels?
  - Are all image pixels **equally important** for learning?
  - Can we learn an **effective representation** of images that is smaller (or much smaller) than the image?

SUPERVISED LEARNING EXAMPLES

- Convolutional Neural Network (CNN):



- Parameters: ANN + lots of filter weights!

SUPERVISED LEARNING EXAMPLES

- ResNet50 [https://arxiv.org/pdf/1502.01852.pdf]:
  - 152 layers!
  - ~25 million parameters!
- Won the ImageNet 2015 image recognition challenge
- **Surpasses** human recognition performance!

WHY DID DEEP LEARNING SUCCEED?

- **Non-linearity** key to success of neural networks
- Deeper networks allow for **greater flexibility** to learn complex input-label relations
- Excellent **generalisation** ability

WHY DID DEEP LEARNING SUCCEED?

- The **backpropagation algorithm** led to the resurgence of neural networks
- An **iterative approach** to **updating parameters**
- Estimates gradients by **backpropagating errors**

WHY DID DEEP LEARNING SUCCEED?

- **Lots of data** available to learn the parameters of the network
- Availability of powerful **Graphics Processing Units (GPUs)** for training deep networks

CONNECTING DEEP LEARNING TO YOUR WORK

- Do you have **lots of labeled data** and want to make **predictions** on new (unseen) data?

- Do you have data that is **serially** arriving (over time)?

- Do you have **some labeled data** that is **similar** to larger labeled databases?

- Do you want to learn **efficient representations** of your data?

- Do you want to **generate new data samples** from existing samples to augment your data?

A FEW CURRENT TRENDS

- DL for IoT
- Deep Reinforcement Learning
- Explainable AI
- Transformers - Language, Speech, Vision
- Continual or Lifelong Learning

RESOURCES

- Plenty of free online resources!
  - Your favorite LLM
  - http://www.deeplearningbook.org
  - Google Colab is a great place to start coding
  - TensorFlow/PyTorch are popular frameworks